

CMIS 3214- Software Engineering

Mid-Semester Examination

Q 1] What is software engineering?

Compare and contrast custom, generic and embedded software solutions.

As a software engineer, it's essential to uphold certain ethical standards. What specific types of ethics are you expected to follow in this role?

[30 marks]

Q 2| A community pharmacy sells medicines to customers either by prescription or over the counter. Each medicine has a unique ID, name, type (e.g., tablet, capsule, syrup), price, and stock quantity. Some medicines require a prescription, and others do not.

The pharmacy employs several pharmacists, each with a unique employee ID, name, contact number and specific qualifications. Pharmacists are responsible for preparing orders and verifying prescriptions.

Customers can purchase medicines by visiting the pharmacy. Each order is assigned a unique ID and includes the date, customer name, and total amount. An order can consist of multiple medicines with specific quantities, and the system should track the subtotal for each item.

If a medicine in an order requires a prescription, then the system must store the prescription document, which contains the prescription ID, issue date, prescribing doctor's name, and notes. A prescription is linked to one or more medicines.

The pharmacy also maintains a record of suppliers, including their supplier IDs, names, companies, and contact information. Suppliers provide the medicines to the pharmacy. One or more suppliers supply each medicine, and a supplier can provide many medicines.

[Draw a class diagram to include the following]

- . Identify main classes.
- . Identify the Attribute for each class.
- . Determine the relationships [show the multiplicity level within the diagrams]

[40 marks]

Q3| Draw an activity diagram for the scenario below. Make sure to use appropriate symbols.

A customer uses a mobile app or website to order food from a restaurant. The activity begins when the customer logs in or creates an account. Once logged in, the customer browses the menu and selects items to add to the cart .

After finalizing the order, the customer proceeds to checkout. At this stage, the system checks whether the customer is logged in. If not, they are redirected to the login page. Once verified, the system calculates the total price and offers multiple payment methods (credit/debit card, digital wallet, or cash on delivery).

If the payment is successful (for online payment methods), the system confirms the order. If payment fails, the user can either retry the order or cancel it. In the case of cash on delivery, payment is marked as "pending".

After confirmation, the system notifies the restaurant. The restaurant prepares the food and updates the status to "ready for delivery". A delivery person is assigned to pick up the food and update the delivery status (delivered or failed).

If delivery fails (e.g., the customer is unavailable), the restaurant contacts the customer for redelivery or a refund. The process ends when the order is either delivered or cancelled due to failure.

[30 marks]